The question

Every contact number MarinFold publishes comes from a 554-protein eval set whose FoldBench half is the **first 100 rows** of `monomer_protein.csv` — chosen before we understood how much of the eval set our training corpora contained (#213, #225). #232 then trained the #199 recipe from scratch on corpora decontaminated against **all** of FoldBench, which makes the other 234 monomers a real held-out test set for those checkpoints for the first time.

So: cut FoldBench's 334 monomers into three sets, score them, and ask whether the set we have been reporting on was telling us the truth.

- **eval-val (97)** — the natural monomers inside the historical FoldBench-100. - **eval-test (217)** — every other natural FoldBench monomer, never scored here. - **eval-denovo (19)** — every de novo designed FoldBench monomer.

Every protein carries a viral flag, because #241 showed the two strata rank models differently.

What was checked before anything was scored

Decontamination, five links, all verified. All 334 monomer sequences are in #225's decontamination reference byte-for-byte; 131,180 training rows match one of them at the applied rule (≥30 % identity over ≥50 % of the shorter sequence) and **all 131,180 are in the drop list — zero survive**; the published corpora match #225's counts; #232's tokenizer pins those exact counts and those exact bucket prefixes; and both W&B runs read only those caches.

What the rule does not cover, priced rather than described. At the applied gate the highest surviving identity to any of the 334 is 0.299. Relax the coverage requirement to 40 % and essentially every protein has a surviving training relative at ≥30 % identity; with no coverage requirement, 65 of them have one at ≥90 % over a fragment. "Decontaminated at 30 %" means that rule, not "no shared subsequence".

Ground truth rebuilt through one path, with the 126 overlapping frozen units reproduced byte-identically. **One protein excluded**: `8uxt_A` (1,596 residues) has no representable contacts-v1 document at an 8,192-token context.

The evaluation reproduces PR #244

Two of the three checkpoints are the ones #244 scored, and all 97 eval-val proteins are inside #244's universe, so the path is validated against a published reference protein by protein rather than against a tolerance on an aggregate: mean R-precision differs by **−0.0031** (m2-p06) and **−0.0023** (m1-p02) with per-protein r = 0.996 — inside the 0.0023 spread #204 measured for one unchanged checkpoint. 333 units × 3 checkpoints, **0 unfinished rollouts**.

The baseline path has its own exact control: re-scoring 12 published proteins through this experiment's rebuilt ground truth reproduces #213's ESMFold and ESMFold2 numbers with **max absolute difference 0.0**.

The result: eval-val was not flattering us

All-range R-precision, quoted as **eval-val (97) / eval-test (217) / eval-denovo (19)**.

The two decontaminated #232 checkpoints: **m2-p06 0.520 / 0.538 / 0.591** and **m1-p02 0.473 / 0.493 / 0.588**. The contaminated reference, #199's cooldown: **0.589 / 0.613 / 0.619**.

Baselines on the same proteins: Protenix-v2 single-seq 0.263 / 0.265 / 0.835; ESMFold 0.750 / 0.753 / 0.795; ESMFold2 0.802 / 0.792 / 0.864; Protenix-v2 + MSA 0.846 / 0.845 / 0.844. The sequence-KNN null scores 0.584 / 0.582 / 0.066 over the unfiltered corpus and 0.407 / 0.426 / 0.050 over the decontaminated one.

Every predictor scores the same or slightly better on the 217 monomers we had never touched, and the contaminated reference moves the same way and by the same amount as the decontaminated checkpoints: the difference-in-differences is **−0.006** (m2-p06) and **−0.004** (m1-p02), an order of magnitude inside the noise. H2 is not supported — the historical FoldBench-100 was not inflating #199's score through memorised homologs. H1 and H3 hold: every predictor moves by under 0.03 between the two sets.

Two things this changes

The KNN null is the yardstick, and it is corpus-specific. Copying the contacts of the ten nearest training sequences scores 0.582 on eval-test out of the unfiltered corpus and 0.426 out of the decontaminated one — 0.156 of memorisable contact map per protein, over 0.2 for 99 of the 314 natural monomers. Each model clears the null over the corpus it trained on (#232 m2-p06 **+0.112**, #199 cooldown **+0.031**) and falls below the null over the richer one. #199's 0.075 lead is therefore not a measurement of what decontamination costs — the runs are not budget-matched, 290,400 vs 145,199 steps — but it bounds how much of any contacts-v1 score is reachable by memorisation.

Protenix-v2 single-sequence is not a comparator on natural proteins. 0.835 on the 19 designs, 0.265 on the 314 natural monomers. The "parity with Protenix-SS" framing came from an eval set that is three-quarters designed protein; on natural FoldBench monomers both #232 checkpoints beat it by more than 0.27. The real gap is to ESMFold2 (−0.255) and Protenix + MSA (−0.307).

Viral proteins are harder for everything except MSA — eval-test viral vs non-viral: m2-p06 0.465/0.542, #199 cooldown 0.497/0.621, ESMFold2 0.608/0.804, seq-KNN 0.262/0.602, Protenix + MSA 0.812/0.847. #241's finding survives on new proteins, and the gap tracks reachable homology.

Designs are easier — against the right natural set

Earlier work (#213, #226, #241) found de novo designs much easier than natural proteins. Here the design advantage looks small: **+0.054** for #232 m2-p06 and **+0.006** for the #199 cooldown against all of eval-test. Both statements are true, about different comparisons.

The published contrast used exp65's 396 idealised `denovo_pdb` designs against the **homology-filtered** natural set (eval2-natural, where MarinFold scores ~0.31-0.36). exp245's eval-denovo is instead the 19 synthetic monomers FoldBench happens to contain — engineered binders and miniaturised folds, n = 19, interval +/-0.09 — and eval-test is *every* natural monomer, only 23 of 217 of which are under 40 % identity to #199's training sequences.

Split eval-test on that axis and the old pattern returns exactly: against natural proteins with no close training homolog, designs are **+0.177 [+0.044, +0.306]** easier for m2-p06 and identically +0.177 for the cooldown. Designs are much easier than natural proteins we have no homolog for; they are about as easy as natural proteins in general, because most natural proteins have homologs.

Protenix-v2 single-seq is the extreme: **+0.570** designs versus all natural, and flat across the identity split (0.243 vs 0.267). It looked competitive only on an eval set that was three-quarters designed protein.

Viral proteins: the penalty is ordered by homology dependenc

Pooled over the 314 natural monomers (295 non-viral / 19 viral), the drop from non-viral to viral is: seq-KNN over the unfiltered corpus **-0.351**, seq-KNN over the decontaminated corpus -0.229, ESMFold2 -0.170, ESMFold -0.153, #199 cooldown -0.123, #232 m1-p02 -0.100, #232 m2-p06 -0.076, Protenix-v2 + MSA **-0.045**, Protenix-v2 single-seq **-0.002**.

That ordering is the finding: the penalty tracks how much a predictor leans on homology. A KNN lookup is nothing but homology and loses a third of its score. ESM's stems are trained on databases where viruses are thin. Protenix single-sequence has no homology signal to lose, and Protenix + MSA fetches one at inference time. An MSA is what closes the viral gap.

The caveat is size: FoldBench's monomers are 5.7 % viral, 19 in total and only 6 in eval-val, where intervals span about +/-0.15. Only the pooled cell supports an argument. Growing this stratum means sampling recent PDB directly, as #241 recommended.

Use from here

The three sets have different **read budgets**, which is the point of the split.

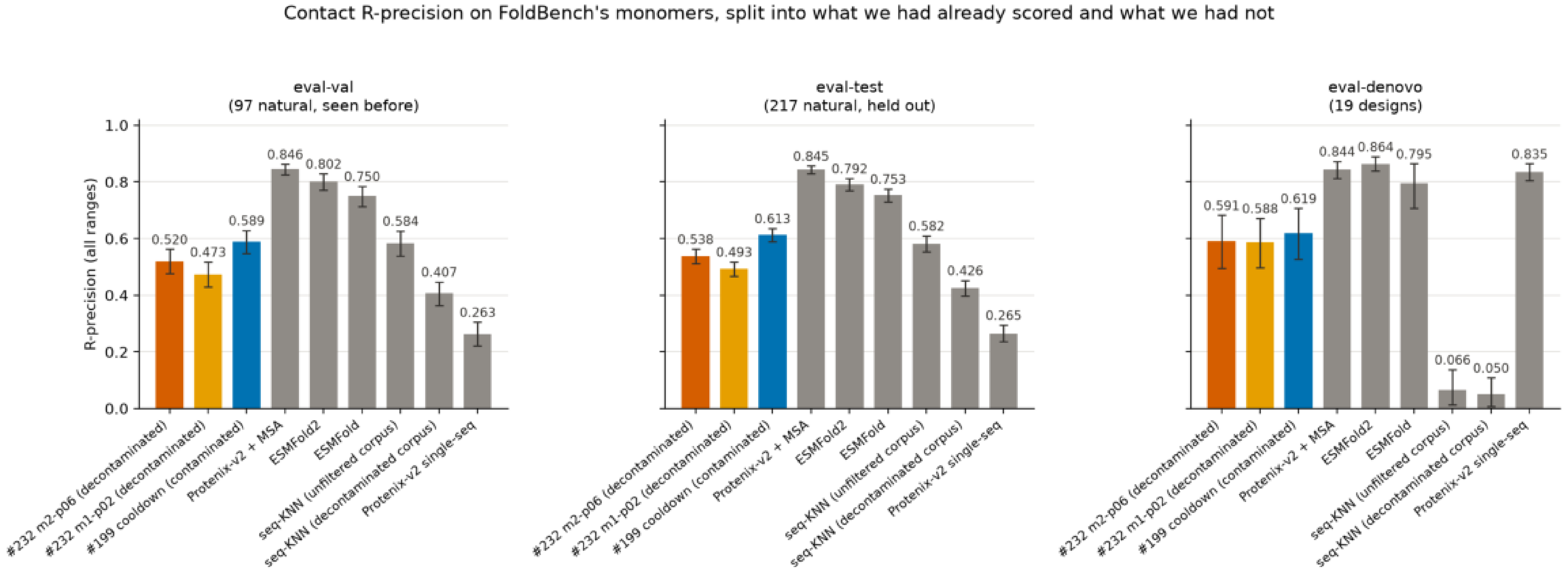
`eval-val` (97) is the working set — checkpoint selection, sweeps, mid-training curves, anything routine. `eval-test` (217) is a held-out confirmation set, read rarely and logged in `data/eval_test_reads.md`; this experiment's construction read is entry 1 and nothing was selected on it. `eval-denovo` (19) keeps designs out of natural-protein means.

The result above is what makes that division workable: every predictor lands within 0.03 of the same number on the two natural sets, so eval-val is an unbiased stand-in for held-out performance and does not need checking against eval-test "just in case". If the read ledger grows a routine tail, eval-test is spent and needs replacing by sampling recent PDB directly (#241).

Everything is on the bucket under `data/contacts-v1-foldbench-monomers-exp245/`.

eval_sets_scoreboard.png

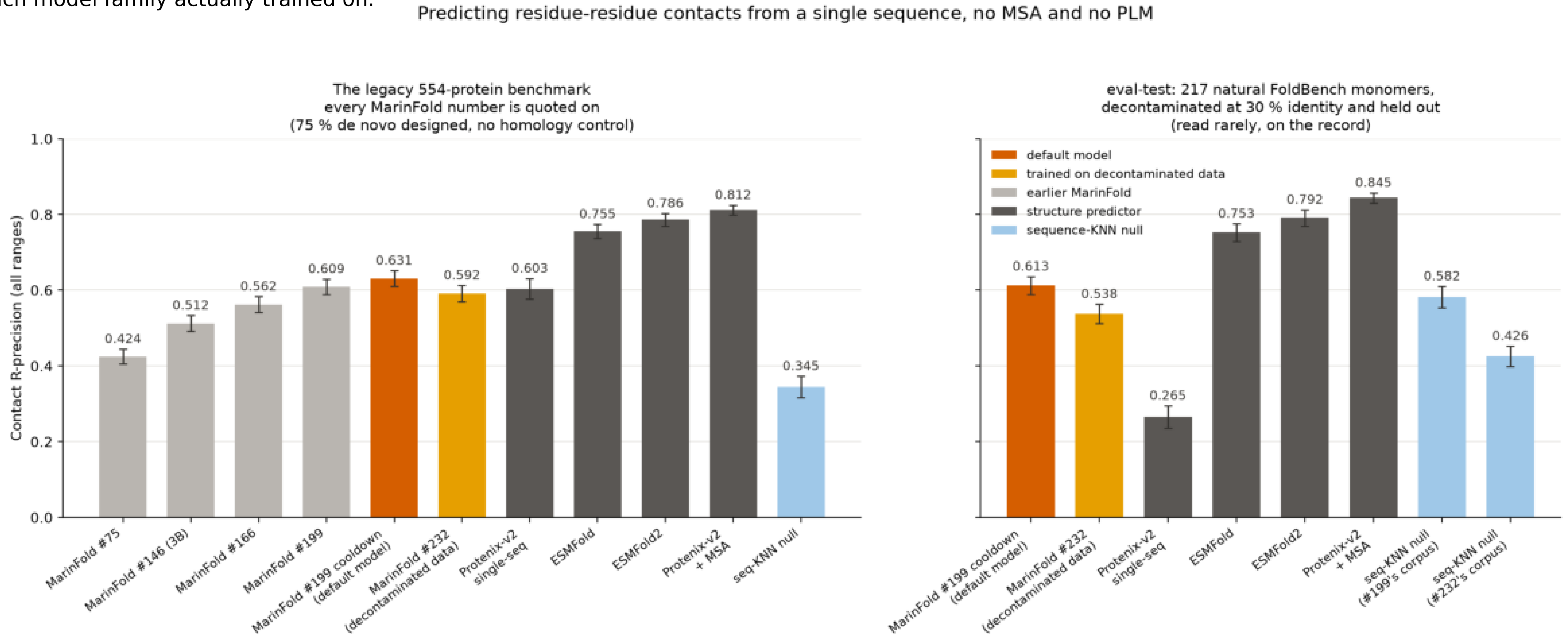
All-range R-precision per predictor on each of the three FoldBench monomer eval sets, with 95 % bootstrap intervals over proteins. MarinFold checkpoints coloured (orange/amber = the decontaminated #232 pair, blue = the contaminated #199 cooldown reference), baselines grey.



$ plot_results.py

readme_performance.png

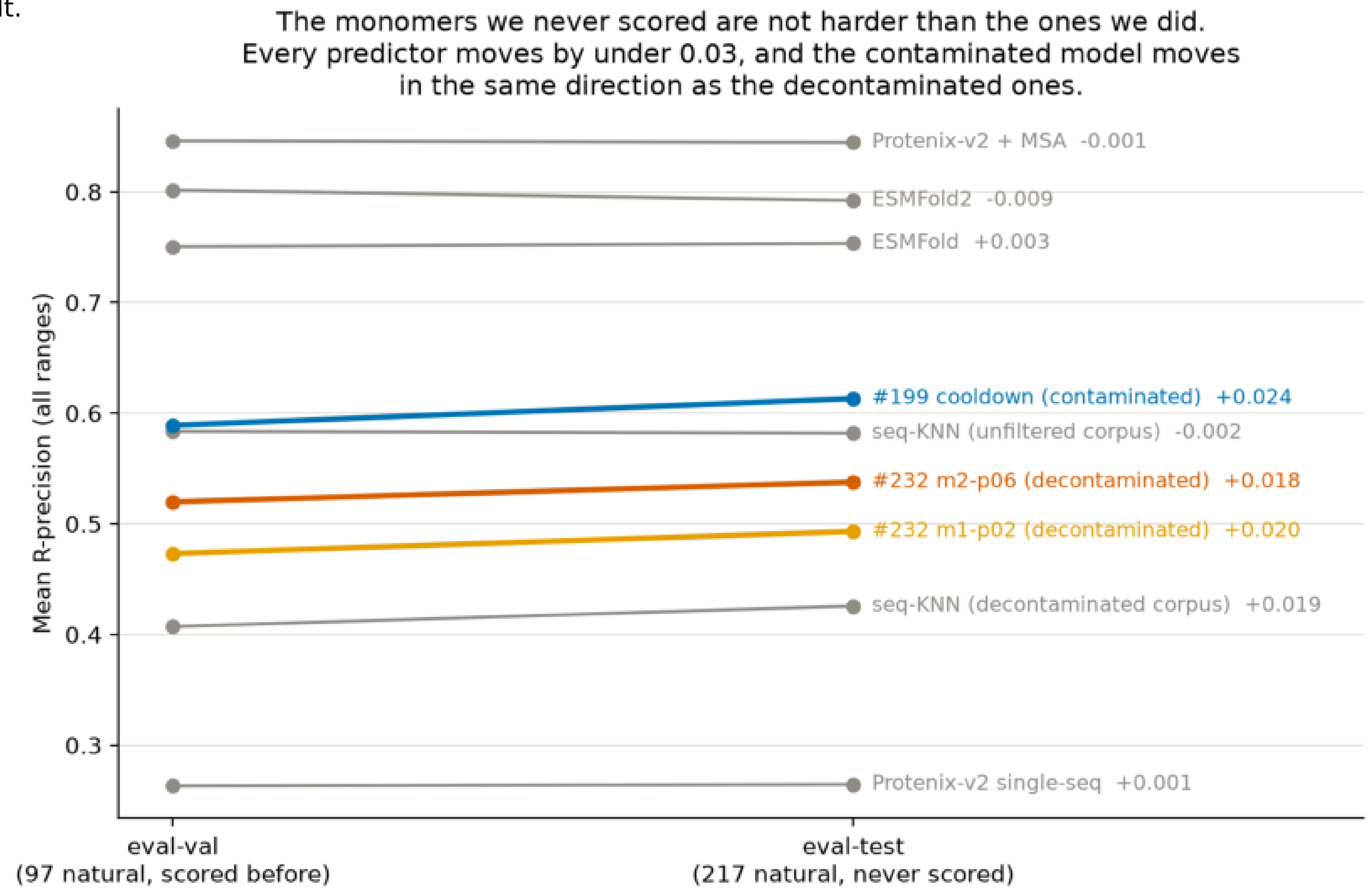
Contact R-precision on the legacy 554-protein benchmark (left, where every published MarinFold number lives) and on exp245's eval-test (right, 217 natural FoldBench monomers held out and decontaminated at 30 % identity). Error bars are 95 % bootstrap intervals over proteins. The two sequence-KNN nulls copy the contacts of each protein's ten nearest training sequences, out of the corpus each model family actually trained on.



$ plot_readme_performance.py

val_vs_test.png

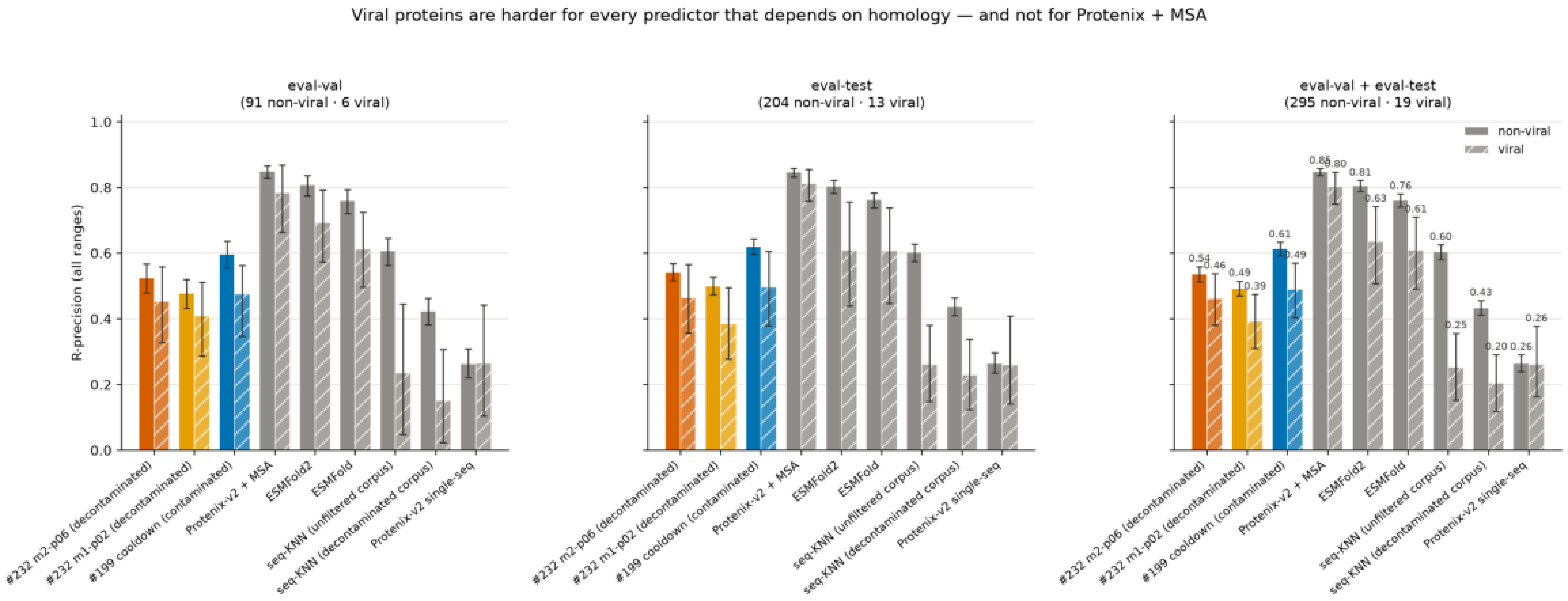
Each predictor's mean R-precision on eval-val (the 97 natural monomers we have always scored) joined to eval-test (the 217 we never had). Every slope is under 0.03 and the contaminated model's is not steeper than the decontaminated ones', which is the experiment's main result.



`$ plot_results.py`

viral_split_scoreboard.png

All-range R-precision per predictor, split viral vs non-viral, on each natural eval set and on the two pooled. Hatched bars are viral. Error bars are 95 % bootstrap intervals; the viral cells hold 6, 13 and 19 proteins, so their intervals are wide by construction. eval-denovo is omitted -- none of the 19 designs is viral.



`$ plot_viral_split.py`